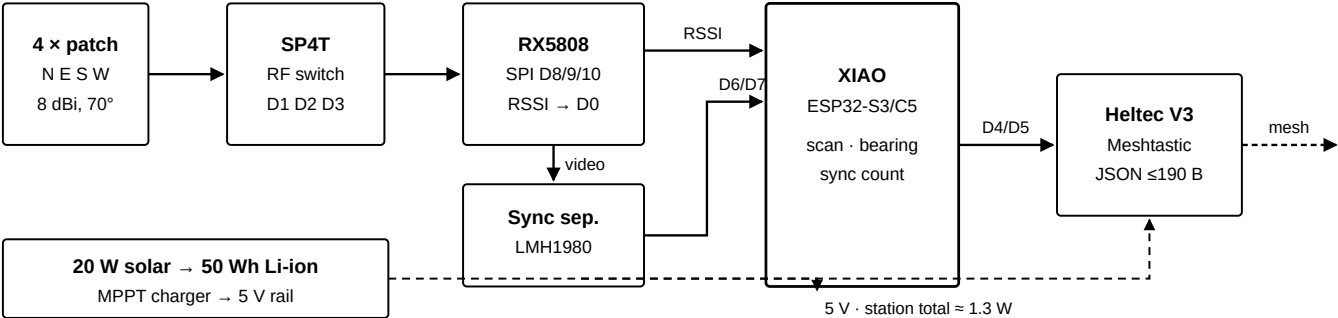


Level 1 Station v2

Sector direction finding and video fingerprinting on the RX5808 analog FPV receiver. Bench build and verification guide.

Document	docs/LEVEL1-V2-HARDWARE.md	Revision	fb6ac6b
Branch	claude/level-1-detection-brainstorm-w8a0vh	Date	2026-09-09
Status	Design — not yet bench validated	Station	

SIGNAL CHAIN AT A GLANCE



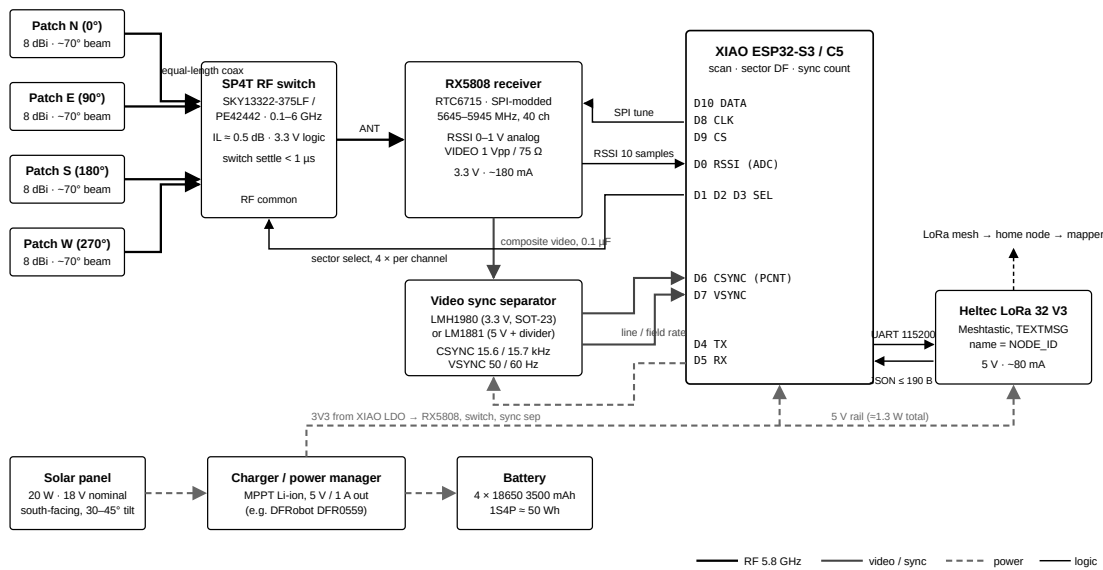
Simplified chain. The full wiring diagram with pin numbers, part numbers and the power tree is on sheet 2.

PARTS, AT A GLANCE

QTY	ITEM	KEY SPEC / PART	HAVE	≈ COST
1	RX5808 receiver	SPI-modded, 5.8 GHz	☐	\$8
4	Patch antenna	5.8 GHz, 8 dBi, ~70°, SMA	☐	\$32
1	SP4T RF switch	SKY13322-375LF / PE42442	☐	\$3
5	u.FL–SMA pigtail	equal length , ≤ 15 cm	☐	\$10
1	Sync separator	LMH1980 (3.3 V) or LM1881	☐	\$3
1	XIAO MCU	ESP32-S3 or ESP32-C5	☐	\$9
1	Heltec LoRa 32 V3	Meshtastic, TEXTMSG 115200	☐	\$20
1	Solar panel	20 W, 18 V nominal	☐	\$25
1	Charger / power manager	MPPT, 5 V / 1 A out	☐	\$15
4	18650 Li-ion	3500 mAh, 1S4P ≈ 50 Wh	☐	\$24
1	IP65 enclosure + brackets	≈200×150×100 mm, 10–15° wedge	☐	\$28
—	Passives	0.1 μF, 10 μF, 100 nF ×4, 2 kΩ ×3, 1000 μF	☐	\$3

Before you start: three values in this guide are unverified and must be measured on the bench, not trusted — the RSSI millivolt-to-dBm calibration points, the XIAO ESP32-C5 D6/D7 GPIO numbers, and the SP4T control truth table. Full BOM on sheet 5, procedure on sheet 6.

1 · Full block diagram



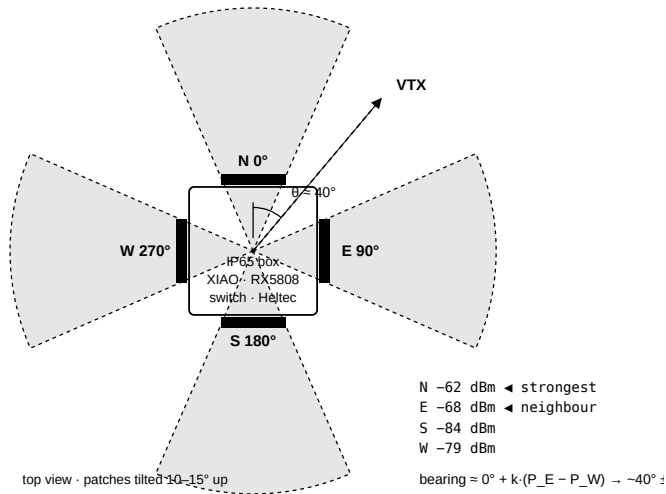
Four patches feed an SP4T switch whose common port replaces the RX5808 whip. The XIAO tunes the receiver over SPI, reads RSSI once per sector, and counts the sync separator's line and field pulses. The XIAO's 3V3 regulator feeds the receiver, switch and sync separator; the 5 V rail feeds the XIAO and Heltec.

WIRING LIST

FROM	TO	XIAO PIN	S3 GPIO	C5 GPIO	NOTES	✓
RX5808 DATA	XIAO	D10	9	10	bit-bang SPI	☐
RX5808 CLK	XIAO	D8	7	8	bit-bang SPI	☐
RX5808 CS	XIAO	D9	8	9	active LOW	☐
RX5808 RSSI	XIAO	D0	1	2	ADC1, 0–1 V	☐
Switch V1	XIAO	D1	2	3	2 kΩ series	☐
Switch V2	XIAO	D2	3	4	2 kΩ series	☐
Switch V3	XIAO	D3	4	5	spare if 2-line decode	☐
RX5808 VIDEO	Sync sep. IN	—	—	—	0.1 μF series, 75 Ω term	☐
Sync sep. CSYNC	XIAO	D6	43	verify	PCNT counter input	☐
Sync sep. VSYNC	XIAO	D7	44	verify	GPIO interrupt	☐
XIAO TX	Heltec RX	D4	5	6	UART 115200	☐
XIAO RX	Heltec TX	D5	6	7	UART 115200	☐
Patches N/E/S/W	Switch RF1–4	—	—	—	equal-length coax	☐
Switch common	RX5808 ANT	—	—	—	shortest run	☐
5 V rail	XIAO 5V, Heltec 5V	—	—	—	1000 μF at Heltec	☐
XIAO 3V3	RX5808, switch, sync sep.	—	—	—	10 μF + 100 nF each	☐
GND	all	—	—	—	single star point	☐

On the ESP32-S3, D6/D7 are the UART0 pins and are free because the console runs over USB-CDC. The C5's D6/D7 GPIO numbers are not recorded in the repository — read them off the board silkscreen and write them in before wiring.

2 · Sector layout and bearing



Top view. Each patch covers about 70°, so an emitter always sits inside one sector and near the edge of a neighbour.

METHOD

Amplitude comparison. Take the strongest sector k and its two neighbours; with powers in dB the bearing is

$$\text{bearing} = az[k] + K \cdot (P[k+1] - P[k-1])$$

K , in degrees per dB, comes from the measured patch pattern. A bench sweep of a VTX around the box every 15° fills the lookup table.

Report bearing_deg and bearing_sigma_deg; sigma grows when the two neighbours are nearly equal, or the peak is weak.

PER-CHANNEL SCAN LOOP

```
tune(channel); delay(TUNE_SETTLE_MS)
for sector in N,E,S,W:
    select(sector)
    delayMicroseconds(200)
    stats[sector] = read_rssi_stats()
hit = max(stats).mean_raw >= THRESHOLD
```

Sweep time $40 \times (30 + 4 \times \sim 2 \text{ ms}) \approx 1.9 \text{ s}$, unchanged in practice. Switching settles in under a microsecond, far inside the 30 ms PLL settle already paid per channel.

VIDEO FINGERPRINT

On the peak channel only, after the sweep: hold the tuner, count CSYNC edges with the PCNT peripheral for 100 ms and time VSYNC edges for 200 ms.

LINE RATE	FIELD RATE	VERDICT	MEANING
15 734 ± 30 Hz	59.94 Hz	NTSC	real analog video carrier
15 625 ± 30 Hz	50 Hz	PAL	real analog video carrier
no stable train	—	none	carrier without video — Wi-Fi, ISM junk, or a bare beacon

sync_q (0–100) is the fraction of 10 ms windows whose count landed within ± 2 of the expected value: a quality figure that does not depend on the RSSI curve at all. The fingerprint fp is <video>/<line_hz>/<freq_peak>, e.g. NTSC/15736/5742, where the peak frequency comes from stepping the synthesizer $\pm 10 \text{ MHz}$ in 2 MHz steps around the hit. Camera crystals and VTX offsets are stable per aircraft over a flight, so two stations reporting the same fingerprint are watching the same drone even when their peak channels differ.

3 · Power budget and output

LOAD	CURRENT	POWER	NOTES
XIAO ESP32-S3/C5, radios off, scanning	~45 mA @ 3.3 V	0.25 W	via the 5 V rail and on-board LDO
RX5808	~180 mA @ 3.3 V	0.60 W	dominant load; cannot be duty-cycled without losing detections
SP4T switch + LMH1980	< 10 mA	0.03 W	the entire DF and fingerprint add-on
Heltec V3, Meshtastic, mostly RX	~80 mA @ 5 V	0.40 W	+ ~120 mA during LoRa TX bursts
Station total		≈ 1.3 W	≈ 31 Wh/day. 20 W panel at 3 peak-sun hours ≈ 60 Wh/day; 50 Wh battery carries ~38 h of darkness

Four separate receivers would have doubled this budget. One receiver behind a switch adds nothing measurable, which is why the switch was chosen.

EXPECTED SERIAL OUTPUT

Detection line on USB at 115200, with the v2 keys appended to the existing contract:

```
{
  "type": "analog_fm",
  "receiver": "rx5808",
  "mac": "AF:00:16:6C:52:03",
  "freq_mhz": 5742,
  "band": "R",
  "ch": 3,
  "rssi_raw": 903,
  "rssi": 903,
  "rssi_mv": 683,
  "rssi_dbm": -66.2,
  "rssi_min": 880,
  "rssi_max": 930,
  "rssi_n": 20,
  "sectors": [-62.1, -68.4, -84.0, -79.2],
  "bearing_deg": 41,
  "bearing_sigma_deg": 14,
  "video": "NTSC",
  "sync_hz": 15736,
  "sync_q": 96,
  "fp": "NTSC/15736/5742",
  "basic_id": "5.8G-R3-5742MHZ",
  "node_id": "RX01",
  "seq": 42
}
```

Heartbeat every 60 s on USB, 120 s on the mesh:

```
{
  "heartbeat": true,
  "node_id": "RX01",
  "receiver": "rx5808",
  "scanning": true,
  "channels": 40,
  "threshold": 600,
  "threshold_dbm": -94.5,
  "temp_c": 41.2,
  "uptime_s": 3600,
  "seq": 42
}
```

The five new keys (sectors, bearing_deg, bearing_sigma_deg, video, sync_hz, fp) are additive. The matching mesh-mapper.py change on branch main must land **before** a station in the field starts emitting them.

RSSI CALIBRATION RECORD

Feed a known level through a step attenuator on a tuned channel, read rssi_mv off the serial monitor, and fill this in. Two points about 50 dB apart set RSSI_CAL_MV_L0/HI and RSSI_CAL_DBM_L0/HI in rx5808.h.

INPUT LEVEL (DBM)	-90	-80	-70	-60	-50	-40	-30
Measured rssi_mv							
Measured rssi_raw							

Noise floor with no transmitter: rssi_raw _____ rssi_mv _____ → set RSSI_THRESHOLD = floor + 200 = _____

4 · Full bill of materials

Single-unit planning prices, quantities per station. Total ≈ \$190 all-in; ≈ \$95 electronics only. The v1 station was ≈ \$45 without power.

RF FRONT END

#	PART	QTY	NOTES	✓	≈ \$
1	RX5808 receiver, SPI-modded	1	Same module as v1. Remove the SPI-disable resistor if the batch ships with it; keep the u.FL pad for the switch feed.	☐	8
2	5.8 GHz patch, 8 dBi, ~70°, SMA	4	Linear is polarisation-agnostic (~3 dB vs an RHCP VTX); RHCP gains 3 dB on the common case, loses ~20 dB on LHCP. Four identical units from one batch.	☐	32
3	SP4T RF switch , 0.1–6 GHz, 3.3 V control	1	SKY13322-375LF or PE42442. ~0.5 dB loss at 5.8 GHz, sub-μs switching. Verify control line count and truth table. Prototype on the vendor eval board (~\$40).	☐	3
4	u.FL–SMA pigtails, equal length ≤ 15 cm	5	Four patches plus switch common. Matched length keeps sector losses matched — this is what the bearing depends on.	☐	10
5	SMA bulkhead feedthroughs	4	One per enclosure face.	☐	8
6	5.8 GHz band-pass filter (<i>optional</i>)	1	Between switch and receiver if a nearby 2.4 GHz access point overloads the front end.	☐	5

VIDEO FINGERPRINT

#	PART	QTY	NOTES	✓	≈ \$
7	TI LMH1980 sync separator (SOT-23-6)	1	3.3 V supply and 3.3 V outputs straight to the XIAO; auto-detects SD/HD. Preferred.	☐	3
7b	<i>alt:</i> LM1881N (DIP-8)	1	5 V part: needs the 5 V rail plus a 2:1 divider or a 74LVC1G17 on each output. Fine for a breadboard.	☐	2
8	0.1 μF coupling cap, 75 Ω termination	1 ea	Per datasheet; the LM1881 also needs 680 kΩ RSET + 0.1 μF.	☐	1

MCU, MESH, POWER, MECHANICAL

#	PART	QTY	NOTES	✓	≈ \$
9	Seeed XIAO ESP32-S3 or ESP32-C5	1	Firmware picks the pinout at compile time.	☐	9
10	Heltec WiFi LoRa 32 V3 + antenna	1	Meshtastic, serial module TEXTMSG 115200, node name = NODE_ID.	☐	20
11	Solar panel 20 W, 18 V	1	South-facing, 30–45° tilt. 10 W covers summer only.	☐	25
12	MPPT charger, 5 V / 1 A out	1	DFRobot DFR0559 or Waveshare Solar Power Manager.	☐	15
13	18650 Li-ion 3500 mAh , 1S4P holder	4	≈ 50 Wh. Protected cells or a BMS board.	☐	24
14	1000 μF low-ESR cap at the Heltec	1	LoRa TX bursts.	☐	1
15	IP65 enclosure ≈ 200×150×100 mm	1	Patches on the four vertical faces at 0/90/180/270°, each tilted 10–15° up.	☐	15
16	Cable gland, desiccant	1		☐	3
17	Pole bracket + four patch wedges	1 set	Face N to true north and record the heading — a mis-oriented box rotates every bearing it reports.	☐	10
18	2 kΩ ×3, 100 nF ×4, 10 μF	—	Switch control series resistors and supply decoupling.	☐	1

5 · Bench procedure

Work top to bottom. Do not move to the next stage until the current one gives the stated result — a bearing built on an uncharacterised switch is not a bearing.

STAGE 1 — RF SWITCH

- ☐ Eval board powered, one patch on RF1, RX5808 on the common port, XIAO GPIOs D1/D2/D3 to the control lines through 2 kΩ.
- ☐ Drive each control combination and record which RF port goes through. **Write the truth table here:**
V1V2V3 = 000 → RF__ 001 → RF__ 010 → RF__ 011 → RF__
- ☐ With a VTX on a known channel, scope the RSSI line while toggling sectors. **Expected:** settled within 1 ms. Measured: _____ μs
- ☐ Compare RSSI with the switch in circuit against the patch wired direct. **Expected:** ≈ 0.5 dB loss. Measured: _____ dB

STAGE 2 — SYNC SEPARATOR

- ☐ RX5808 VIDEO → 0.1 μF → separator input, 75 Ω terminated. Scope CSYNC with a real VTX on the tuned channel. **Expected:** clean train at 15.6–15.7 kHz. Measured: _____ Hz
- ☐ Scope VSYNC. **Expected:** 50 Hz (PAL) or 59.94 Hz (NTSC). Measured: _____ Hz
- ☐ Tune to a channel carrying only Wi-Fi (5745–5825 MHz band). **Expected:** no stable sync train. Confirmed: yes / no
- ☐ Confirm output logic levels reach the XIAO's input threshold (LM1881 at 5 V needs the divider).

STAGE 3 — RSSI CALIBRATION

- ☐ Fill the calibration table on sheet 4 using a step attenuator, or a VTX at two known open-field distances.
- ☐ Enter the two points in rx5808.h as RSSI_CAL_MV_L0/DBM_L0 and RSSI_CAL_MV_HI/DBM_HI, then reflash.
- ☐ Set RSSI_CAL_OFFSET_DB per station so a fleet agrees on the same source. Value used: _____ dB
- ☐ Confirm the heartbeat now reports a sane threshold_dbm (expect roughly -90 to -95). Reported: _____ dBm

STAGE 4 — PATTERN CALIBRATION

- ☐ Mount all four patches on the box, VTX at 30 m on a known bearing, clear line of sight.
- ☐ Rotate the box in 15° steps through a full turn, logging all four sectors values at each step.
- ☐ Derive K in degrees per dB from the neighbour difference. Value: _____ °/dB
- ☐ Save the sweep to docs/ as the station's bearing calibration.

STAGE 5 — FIELD ACCEPTANCE

- ☐ Two v2 stations 300–500 m apart, both positions set in the mapper, both named to match their Meshtastic nodes.
- ☐ Walk a VTX between them on ten marked points. **Pass:** the 90 % circle contains the true position on at least 9 of 10. Score: ____ / 10
- ☐ **Pass:** video reads NTSC or PAL on every report, never none, for a live VTX.
- ☐ Leave both stations on solar for 72 h. **Pass:** no brownout, heartbeat gap never exceeds 5 min.

Built by		Date	
Verified by		Date	
Station ID		True-north heading of face N	